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Methodological challenges in explainable AI for fraud detection: a systematic literature review

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Abstract

As complex black-box Artificial Intelligence (AI) models become integral to high-stakes domains like fraud detection, the need for transparency is becoming critical. Explainable AI (XAI) aims to make algorithmic decisions understandable to stakeholders. While systematic reviews have mapped the use of XAI in the broader financial sector, a focused synthesis of the methodological challenges unique to fraud detection remains a critical gap. This systematic literature review addresses this gap, following PRISMA 2020 guidance and synthesizing 49 peer-reviewed studies (2021–2025) to identify current practices and foundational challenges. Our analysis reveals that while post-hoc methods like SHAP and LIME are prevalent, their application is undermined by two systemic methodological flaws that threaten the validity of current research. First, we identify an explainability–imbalance paradox, where common data resampling techniques used to manage class imbalance can distort background distributions or local neighborhoods and thereby compromise the faithfulness and fidelity of post-hoc explanations. Second, we discover a profound evaluation vacuum, with around 80% of the analyzed studies using model predictive performance as a proxy for explanation quality rather than directly evaluating the explanations themselves. We also critically assess when post-hoc explanations are appropriate versus when intrinsically interpretable models are preferable in auditable and high-risk contexts. Based on these findings, we propose an explanation-aware training and evaluation checklist and a research agenda to guide the field toward standardized, explanation-level evaluation and the development of imbalance-robust, explanation-preserving data-processing methods.

Keywords Explainable artificial intelligence (XAI) · Fraud detection · Systematic literature review · Methodological challenges · Class imbalance · Evaluation metrics.

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1 Introduction

1.1 The "black box" dilemma in AI-powered fraud detection

The global financial and healthcare systems face a persistent and escalating threat from sophisticated fraudulent activities, leading to substantial economic losses and compromised patient care. In response, institutions have increasingly turned to advanced Artificial Intelligence (AI) and Machine Learning (ML) models, such as deep neural networks (DNNs) and gradient boosting machines, which demonstrate superior predictive power in identifying subtle, complex, and evolving fraudulent patterns that often elude traditional rule-based systems (Mozumder et al. 2025; Lee et al. 2025). These models are capable of processing large, high-dimensional datasets to uncover subtle patterns of fraud from single anomalous transactions to complex multi-stage billing schemes (Samek et al. 2017).

However, this enhanced performance often comes at a significant cost: transparency. The very complexity that makes these models powerful also renders their internal decision-making processes opaque and inscrutable to human operators. This "black box" problem is a fundamental challenge in the deployment of modern AI (Bennetot et al. 2025). When a model flags a transaction as fraudulent or denies a claim, stakeholders, including fraud analysts, compliance officers, customers, and regulators, are often left without a clear, understandable reason for the decision. This inability to interpret a model's logic creates unacceptable risks in high-stakes environments, posing significant barriers to trust, accountability, and regulatory compliance (Lin et al. 2025). Throughout this study, we use faithfulness to mean the degree to which an explanation reflects the model's true decision behavior, and fidelity to mean the agreement between an explanation (or surrogate) and the model's outputs.

1.2 The imperative for transparency: trust, accountability, and regulatory drivers

Explainable AI (XAI) has emerged as a critical field of research and practice dedicated to resolving this transparency deficit. The primary goal of XAI is to develop a suite of techniques that can make the decisions of complex models understandable to their human users, transforming them from opaque oracles into transparent partners in the decision-making process (Gao et al. 2024; Bennetot et al. 2025). By providing insights into which features drove a prediction and how, XAI aims to build trustworthy, auditable, and fair AI systems (Zhou et al. 2023).

Several converging forces are accelerating the need for explainability. Operationally, fraud analysts need to understand a model's reasoning to validate its alerts, reduce false positives, and conduct effective investigations (Vorobyev and Krivitskaya 2022). Ethically, there is a growing demand for algorithmic accountability, ensuring that automated decisions are fair and justifiable. Critically, regulatory bodies worldwide are beginning to mandate transparency. Frameworks like the European Union's General Data Protection Regulation (GDPR) allude to a "right to explanation," requiring that individuals be given meaningful information about the logic involved in automated decisions (Sambasivan et al. 2021). In sectors like finance and healthcare, where the consequences of an erroneous or biased decision can be severe, explainability is not merely a desirable feature but a fundamental prerequisite for responsible and legal deployment (Capuano et al. 2022). Importantly, in auditable, consequential decisions (e.g., chargebacks or AML alerts), we argue that post-hoc rationales may be insufficient unless their stability and faithfulness are empirically verified. Therefore, fundamentally interpretable models with domain constraints can provide stronger compliance assurance.

1.3 Research questions and contributions

This systematic review is structured to answer five key research questions (RQs) that address the current state, challenges, and future of XAI in fraud detection. These questions form the narrative backbone of this manuscript, with each subsequent section dedicated to addressing them through a synthesis of the evidence from the 49 selected studies:

RQ1 What are the dominant XAI techniques and frameworks being applied to fraud detection models?

RQ2 What are the primary application domains for XAI in fraud detection, and what challenges do they face?

RQ3 How does class imbalance affect the generation and reliability of explanations?

RQ4 What methodologies are used to evaluate the quality, stability, and utility of the generated explanations?

RQ5 What are the key research gaps and most promising future directions for the field?

By addressing these questions, this review makes several novel contributions to the field. It provides a focused, systematic, and critical synthesis of the methodological challenges of XAI for fraud detection. It identifies and articulates a critical, under-discussed challenge termed the explainability-imbalance paradox, where common data preprocessing steps may inadvertently distort the reference distributions or perturbation neighbourhoods used by explainers and thereby degrade explanation faithfulness/fidelity. Unlike broader finance-wide XAI reviews, our scope uniquely concentrates on the fraud detection sub-domain, where the interplay of explainability with extreme class imbalance, adversarial dynamics, and specific evaluation needs creates a distinct set of methodological challenges. Finally, it issues a call to action to address the "evaluation vacuum", the profound lack of standardized methods for validating explanation quality. It proposes a progressive research agenda to guide the field toward greater development and real-world impact.

To contextualize the scope of this review, we propose a conceptual framework outlining the role of XAI in a typical fraud detection workflow (Fig. 1). The framework begins with diverse data sources, such as transaction records, insurance claims, and KYC documentation, which are processed and transformed into model-ready features. These feed into the fraud detection model, often a complex machine learning or deep learning system, whose outputs are subsequently interpreted through an XAI layer (e.g., SHAP, LIME, or counterfactual explanations). The generated explanations support human-in-the-loop decision-making, enabling analysts to audit and respond to model predictions. Feedback from both decisions and explanations informs iterative model refinement, ensuring that the detection system remains accurate, interpretable, and aligned with regulatory requirements. This

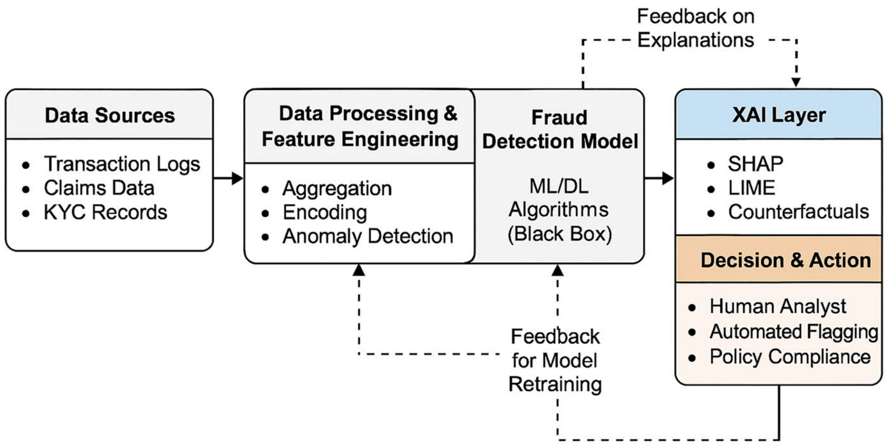

Fig. 1 Conceptual framework for integration of XAI within a fraud detection workflow

framework serves as the analytical lens through which the selected studies are synthesized in the following sections. The subsequent methodology and analysis adopt this framework as a guiding structure for synthesizing findings from the selected studies.

2 Related work

2.1 Overview of existing systematic reviews

The existing literature contains several systematic reviews focusing on either XAI or fraud detection independently, but very few that integrate both. Reviews on XAI often provide broad taxonomies of techniques and survey their application across diverse fields, such as finance or healthcare, but without a deep focus on the unique challenges of fraud detection, including adversarial behaviors, regulatory auditability requirements, and extreme class imbalance. For instance, a review by (Martins et al. 2024) provides a broad taxonomy of XAI applications in finance but does not systematically analyze fraud-specific implementations or evaluate whether the explanations are stable, faithful, or practically useful. On the other hand, systematic reviews of fraud detection frequently survey ML and DL techniques, comparing their predictive performance on benchmark datasets (Niaz et al. 2024). These reviews often acknowledge the transparency problem as a limitation, yet they do not evaluate how XAI methods are deployed, how their explanations are validated, or whether post-hoc approaches are even appropriate in high-risk contexts (Pourhabibi et al. 2020; Chen Yisong et al. 2024; Mohsin and Nasim 2025). This lack of explanation-level scrutiny has created a persistent knowledge gap between these two critical fields.

While our quantitative synthesis focuses on the 2021–2025 period, our critical analysis explicitly evaluates the foundational principles established in seminal, earlier literature. We position the works of (Doshi-Velez and Kim 2017) on evaluation and (Barredo Arrieta et al. 2020) on taxonomy as the theoretical bedrock against which we measure recent progress, or the lack thereof. This reframing transforms the limited timeframe from a potential weakness into a methodological strength. It allows us to construct a more powerful argument: making our review a “report card” on how well the applied field has responded to these foundational challenges to action, revealing a significant gap between established theory and recent practice. In fraud-specific contexts, early anomaly detection reviews, e.g., (Chalapathy and Chawla 2019) explored unsupervised techniques that implicitly touch on explainability, particularly when decisions must be interpreted for security audits or regulatory reporting. These early insights provide essential theoretical anchors for assessing incompleteness and inconsistency of current practices.

2.2 Gaps in explainability-focused fraud detection reviews

The few existing surveys that touch upon XAI in finance or cybersecurity tend to be high-level overviews rather than systematic reviews. They often discuss the importance of SHAP and LIME, but do not provide a structured analysis of their application contexts, the types of models they are used to explain, the background distribution or perturbation design required for faithfulness, or the methods used to evaluate the explanations themselves (Capuano et al. 2022; Benito et al. 2024). This leaves a critical gap for a review that systematically maps which XAI techniques are used in fraud detection and how they are applied and validated. The novelty of this study lies in its shift from a descriptive review (“what techniques are used?”) to a critical methodological audit that asks: “why is the current approach failing to produce faithful and reliable explanations?” and “what foundational problems are being ignored?”. As shown in Table 1, our study is positioned to fill this gap by providing an integrated, systematic analysis of the intersection between XAI and fraud detection.

3 Methodology

We conducted a PRISMA 2020–guided systematic literature review (Page et al. 2021) to map and critically synthesize research on explainable AI (XAI) for fraud/crime detection. The workflow comprised: (i) database search and two-stage deduplication, (ii) multi-stage screening (Title/Keyword → Abstract → Full-text) against predefined eligibility criteria, (iii) standardized data extraction using a piloted codebook, and (iv) quality/risk-of-bias appraisal, followed by a narrative synthesis supported by descriptive statistics (with 95% CIs). Inter-rater reliability at full text was Cohen’s $\kappa = 0.82$, and the disagreements were resolved by consensus. To ensure reproducibility, all complete database search strings and final search dates are provided in the Appendix. In addition, the full screening log, data-extraction codebook, row-level extraction dataset, risk-of-bias scores, and analysis notebooks are openly available in our OSF repository at https://osf.io/9nykd/.

3.1 Search strategy

A systematic and comprehensive search strategy was designed to identify all relevant peer-reviewed literature at the intersection of explainable AI and fraud detection. We searched Scopus, Web of Science (Core Collection), and IEEE Xplore published between January 1, 2021, and June 2025. Queries combined three concept blocks: (1) explainabil-

Table 1 Comparative analysis of systematic reviews in XAI and fraud detection

Study	Scope & Focus	Methodology	Key Findings/Contribution	Identified Gaps
Doshi-Velez and Kim (2017)	Foundational critique of XAI evaluation	Conceptual framework proposes taxonomy of evaluation strategies	Highlights need for functionally- and human-grounded evaluation standards.	Lacks application-specific XAI recommendations. This study extends these principles to the fraud detection domain
(Pourhabibi et al. 2020)	Fraud Detection	Review of graph-based anomaly detection methods.	Provides a survey of graph-based models for fraud detection.	Does not systematically cover XAI techniques for interpreting these complex graph models. This study directly addresses the interpretation layer.
(Černevičienė and Kabašinskas 2024)	XAI in Finance	SLR of 138 articles (2005–2022).	Identifies popular financial tasks (credit, stock prediction, fraud) and dominant XAI methods (SHAP, feature importance, rule-based).	Primarily a descriptive mapping of techniques and applications. It does not critically analyze the interaction of these methods with data-specific challenges like imbalance or the rigor of their evaluation. This study focuses on that critical analysis.
(Mohsin and Nasim 2025)	XAI in Finance	Bibliometric and content analysis of 323 papers.	Maps research trends, topic clusters, and popular techniques (SHAP, attention mechanisms).	Provides a high-level, quantitative trend analysis. It lacks the deep, qualitative synthesis of specific methodological flaws and their implications, which is the core of this study.
(Chen Yisong et al. 2024)	Deep Learning for Fraud Detection	SLR (Kitchenham framework) of 57 studies (2019–2024).	Surveys the application and performance of DL models (CNN, LSTM, Transformers) in financial fraud detection.	Explicitly acknowledges the “black box” problem as a limitation but does not systematically review or analyze XAI solutions. This reinforces this study’s premise that a dedicated review on the XAI aspect is needed.
(Barredo Arrieta et al. 2020)	General XAI	Foundational/ Tutorial Paper.	Provides influential taxonomies of XAI concepts, methods (ante-hoc vs. post-hoc), and goals (e.g., trust, fairness).	Foundational and conceptual, not an empirical review of a specific application domain. This study applies these concepts to empirically analyze the state of XAI in the fraud detection domain.

Table 1 (continued)

Study	Scope & Focus	Methodology	Key Findings/Contribution	Identified Gaps
This Study	XAI in Fraud Detection	PRISMA-guided SLR of 49 studies (2021–2025); extraction dataset + codebook + analysis notebooks publicly released (OSF/Zenodo) for reproducibility.	Moves beyond taxonomy to critically analyze the validity of current practices. Formally defines the "explainability-imbalance paradox" and the "evaluation vacuum." Introduces an explanation-aware checklist.	**Contribution**: Fills the identified gaps by providing a focused, systematic, and critical synthesis of methodological challenges hindering the reliable deployment of XAI in fraud detection. Diagnoses systemic issues (faithfulness/fidelity failures, instability, proxy-only evaluation) and proposes a rigorous, reproducible research agenda.

ity (e.g., "explainable artificial intelligence", XAI, SHAP, LIME, counterfactual*, attention mechanism*), (2) fraud, crime, anomaly (e.g., fraud*, "fraud detection", AML, illicit transactions), and (3) machine learning (e.g., "machine learning", "deep learning", neural network*). Database-specific field tags and filters (journals-only) were applied. The exact query strings, filters, and per-database search dates are provided in the Appendix, and the pre-deduplication counts are provided in our OSF registration (see Data/Code Availability). This timeframe, 2021–2025, was chosen to capture the most recent advancements in this rapidly evolving field. Recent bibliometric studies indicate a surge in research activity at the intersection of XAI and fraud beginning in 2021, ensuring that the review reflects the current state of the art.

3.2 Study selection and screening process

The selection of studies for this review followed the structured, multi-stage process detailed in the PRISMA 2020 flow diagram (Fig. 2). After the initial database search and the automatic removal of duplicates, a two-stage manual screening process was conducted by two independent reviewers. After export from the three databases, duplicates were removed using a two-stage procedure: EndNote automated deduplication (DOI/normalized title) followed by manual verification. We then performed Title/Keyword screening, Abstract screening, and Full-text eligibility assessment against predefined criteria (Table 2). Two reviewers screened independently at each manual stage; disagreements were reconciled by consensus ($\kappa = 0.82$ at full text). Full-text exclusions were recorded with a primary reason; the complete exclusion table is available at OSF. This detailed assessment ensured that only primary empirical studies of sufficient quality and relevance were included in the final synthesis. Any disagreements between the reviewers regarding the inclusion of a study were resolved through consensus discussion.

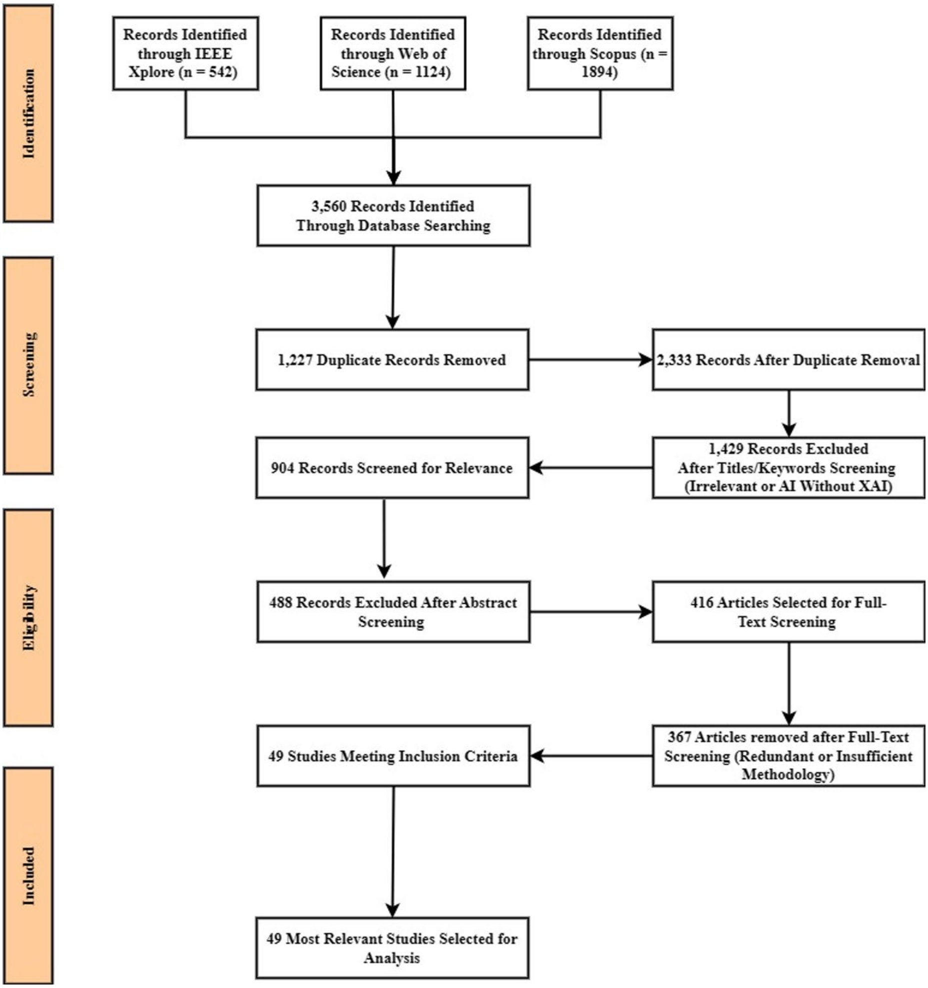

Fig. 2 Study selection flow diagram following PRISMA guidelines

3.3 Establishing rigor: a PRISMA-Guided review of the field

While several surveys on XAI have been conducted, they often provide broad overviews of techniques across many domains (Gao et al. 2024; Bennetot et al. 2025). Conversely, reviews of AI in fraud detection frequently acknowledge the black box problem do not systematically analyze the adoption, validation, or impact of XAI solutions, nor do they assess explanation fidelity, faithfulness, or stability. This has created a clear and evidenced knowledge gap at the critical intersection of these two fields. This study aims to fill that gap by conducting a systematic literature review (SLR) to provide a focused, comprehensive, and critical synthesis of the state-of-the-art in XAI for fraud detection (Fig. 1).

This study adheres strictly to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al. 2021) to ensure a transparent, rigorous, and replicable review process. As shown in the PRISMA flow diagram (Fig. 2), the

Table 2 Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Studies published between 2021 and 2025, written in English, and accessible in full text.	Papers irrelevant to fraud or anomaly detection (e.g., generic credit scoring, sentiment analysis, or unrelated cybersecurity).
Papers explicitly reporting at least one explainability component, such as SHAP, LIME, counterfactual reasoning, attention mechanisms, feature attribution, rule extraction, or intrinsically interpretable models.	Studies focusing solely on AI-driven fraud detection without addressing explainability (no interpretable or post-hoc explanation reported).
Empirical studies that evaluate interpretability, faithfulness, fidelity, stability, or human-centered utility alongside fraud detection performance.	Duplicate entries, extended abstracts, workshop proposals, or redundant studies lacking novel contributions.
Research providing comparative insights, explanation-level validation, or visualization to support model interpretability in fraud detection.	Papers with insufficient methodological details, unverifiable results, missing evaluation data, or unavailable full text (registered on OSF).

initial search spanned three major academic databases; IEEE Xplore ($n=542$), Web of Science ($n=1{,}124$), and Scopus ($n=1{,}894$), yielding 3,560 records in total. Deduplication was conducted in two stages: (i) automated removal via EndNote (1,227 duplicates), and (ii) manual verification by two reviewers to ensure no metadata inconsistencies remained. After deduplication, 2,333 unique records proceeded to title and keyword screening, resulting in the exclusion of 1,429 irrelevant records. Next, 904 abstracts were screened, with 488 records excluded for not meeting relevance or explainability criteria. The remaining 416 full-text articles were then assessed against the formal inclusion/exclusion criteria (Table 2), and a complete exclusion log with reasons is provided in the dataset registered on OSF.

To ensure screening consistency, two reviewers independently assessed all full-text articles. Inter-rater reliability was calculated using Cohen's kappa ($\kappa=0.82$), indicating "almost perfect" agreement and confirming the robustness of our selection procedure. Following full-text eligibility assessment, 367 articles were excluded due to redundancy, insufficient methodological detail, missing explanation components, unverifiable results, or failure to meet minimum inclusion criteria. This rigorous multi-stage process resulted in a final corpus of 49 peer-reviewed studies for in-depth qualitative and methodological analysis.

3.4 Data extraction and quality assessment

For each of the 49 included journal articles, we used a structured data-extraction codebook to ensure consistent coding. The following fields were extracted:

- Bibliographic metadata: Authors, Year, Title, Journal.
- Application domain: e.g., Healthcare, Financial Services (banking/credit card), Insurance, E-commerce, AML/Crypto.
- Specific task: e.g., card-present/card-not-present, claims fraud, illicit transactions, mer-

chant risk, account takeover.
- AI/ML model explained: e.g., XGBoost, Random Forest, DNN/CNN/RNN, Logistic Regression, GBT.
- XAI paradigm & method: e.g., feature attribution (SHAP, LIME, IG), counterfactual, interpretable/rule-based models (e.g., GAM/EBM), global plots (PDP/ICE/ALE).
- Imbalance handling technique: data-level (SMOTE/variants, undersampling, augmentation) or algorithm-level (class weights/cost-sensitive); None if not used.
- Evaluation method category (mutually exclusive): Proxy, Functionally-Grounded, or Human-Grounded.
- Reported explanation checks/metrics: e.g., fidelity/faithfulness tests, stability/robustness, deletion/preservation, sensitivity analyses.
- Data source: public benchmark vs. proprietary/production.
- Extraction procedure. Two reviewers independently extracted all items using the piloted codebook; discrepancies were resolved by consensus with reference to the source article, yielding a single reconciled dataset.

Quality (risk-of-bias) assessment: We applied a domain-appropriate rubric scoring five criteria: Clarity of objectives, 2) Methods transparency, 3) Dataset description, 4) Explanation evaluation adequacy (e.g., fidelity/stability checks), and 5) Imbalance handling. Two reviewers coded all 49 studies independently with consensus reconciliation. Summary scores and the Risk-of-Bias figure are provided in the Appendix (with details for each study available in the OSF repository).

4 Results

This section presents the core findings from the qualitative synthesis of the 49 selected studies. It begins with a descriptive analysis to provide a high-level overview of the research landscape and a detailed thematic analysis organized by the guiding research questions.

4.1 Descriptive analysis of the literature

The bibliometric data extracted from the 49 articles reveal a rapidly growing field needing attention. As shown in Fig. 3, the number of publications on XAI for fraud detection has increased steadily year-over-year, with a pronounced surge from 2023 onwards. This trend indicates that the convergence of mature XAI toolkits, accessible high-performance computing, and increasing regulatory pressure has created a fertile ground for research and innovation.

The application of XAI in this context is not evenly distributed but is instead concentrated in a few high-stakes domains where the need for transparency is most acute. As illustrated in Fig. 4, Financial Services is the predominant application area encompassing problems like credit card fraud, loan default prediction, and illicit transaction monitoring. This is followed by healthcare, where XAI is applied to tasks such as medical claims fraud and clinical diagnosis. A smaller but significant portion of research is dedicated to emerging areas like Blockchain security and other industrial applications. This distribution directly reflects the

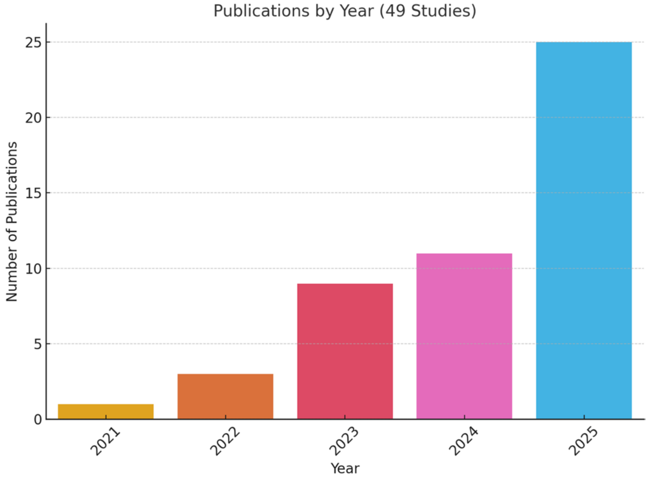

Fig. 3 Publications by year (2021–2025)

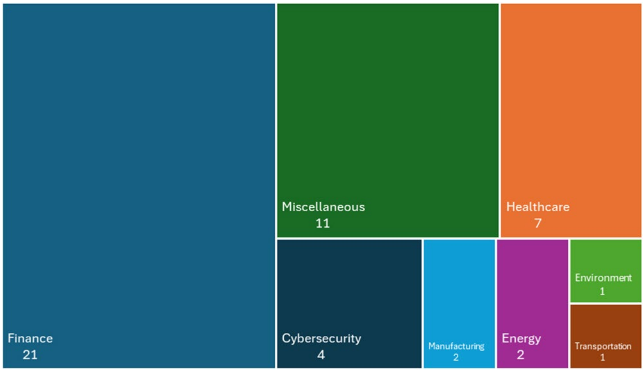

Fig. 4 Application domains in selected studies

domains where the financial, social, or human cost of an incorrect and unexplainable AI decision is highest.

A clear preference has emerged for a small set of post-hoc explanation methods, which are often applied to a mix of inherently interpretable models and more complex black-box systems. Figure 5 shows that SHapley Additive exPlanations (SHAP) is by far the most widely used XAI method in the reviewed literature, appearing in 26 of the 49 studies. It is followed by Local Interpretable Model-agnostic Explanations (LIME), which was used in 9 studies. Decision Trees (7), Random Forests (7), and XGBoost (6) also feature prominently as explainable model approaches. Other techniques, such as Transformer models, Counterfactual explanations, and Convolutional Neural Networks (CNNs), each appear in 5 studies, while a range of methods, including LightGBM, Support Vector Machines, Logistic Regression, LSTM, Autoencoders, DICE, and Graph Neural Networks (GNNs), have more limited adoption.

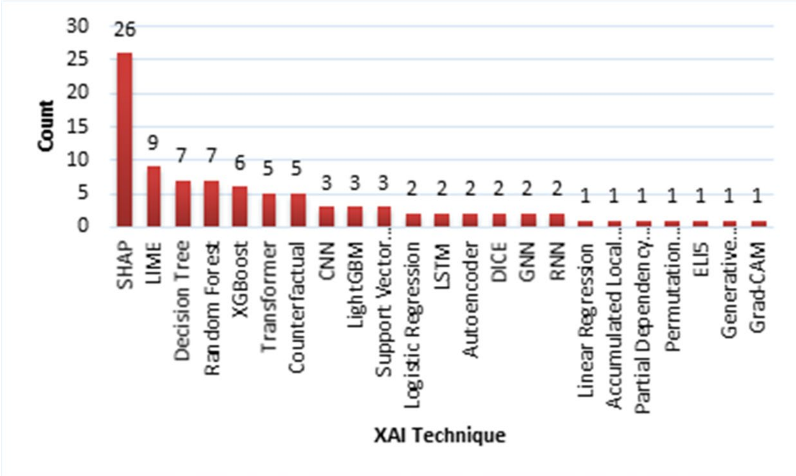

Fig. 5 Distribution of XAI methods and model types analyzed in the literature

4.2 The methodological landscape of XAI in fraud detection (RQ1)

4.2.1 The dominance of Post-Hoc, model-agnostic explanations

The analysis reveals that post-hoc, model-agnostic explanation methods overwhelmingly dominate the current methodological landscape. These techniques are popular for their flexibility, as they treat the underlying prediction model as a black box and can be applied to any pre-trained model without changing its architecture. This pragmatic decision demonstrates a need to add explainability to existing, high-performing fraud detection systems. However, this has led to an overreliance on a limited set of techniques, which has significant downstream effects and creates conditions that increase the risk of fidelity issues, as discussed later in this review.

The review shows that SHapley Additive exPlanations (SHAP) is the dominant method, applied in over 50% of the reviewed studies. This prevalence is not arbitrary but is driven by a confluence of factors that make it particularly suitable for high-stakes applications. First, its strong theoretical grounding in cooperative game theory provides a desirable property of fair attribution that appeals to the academic community and practitioners seeking robust, defensible justifications. Second, the availability of mature, well-documented open-source toolkits has lowered the barrier to entry, driving widespread adoption. The critical implication of this dominance, and the key to fitting these pieces together, is its direct alignment with the needs of the most prominent application domain: financial services. The intense regulatory pressure for auditable decisions makes a theoretically sound method like SHAP an invaluable tool for compliance. Its applications span nearly every domain identified, from explaining complex deep learning frameworks for Medicare fraud detection(Awosika et al. 2024; Mozumder et al. 2025) and identifying key drivers of automobile loan risk(Mayrhofer and Filzmoser 2023; Lin et al. 2025) to interpreting unsupervised anomaly detection models like autoencoders (Antwarg et al. 2021).

Local Interpretable Model-agnostic Explanations (LIME) is the second most common technique, valued not just for its ease of implementation but for its specific utility in operational workflows. LIME operates on a more intuitive principle: to explain a single prediction, it creates a simple, inherently interpretable surrogate model (e.g., a linear model or decision tree) that is locally faithful to the complex model's behavior in the vicinity of that prediction (Doshi-Velez and Kim 2017; Martins et al. 2024). This directly mirrors the task of a fraud analyst who must investigate one flagged transaction at a time, answering the immediate question: 'Why was this specific case flagged?' This focus on local, case-by-case interpretability explains its frequent application in tasks like credit card fraud detection (Damanik and Liu 2025) and supply chain credit assessment (Shi et al. 2025), even as some studies note its potential for instability across different runs. The dominance of SHAP and LIME constitutes a 'methodological monoculture' born of necessity. High-stakes domains like finance and healthcare cannot easily sacrifice the predictive power of their complex, fine-tuned models. Post-hoc methods offer a pragmatic compromise, promising transparency without requiring a complete overhaul of existing systems. This symbiotic relationship between flexible, post-hoc methods and the inertia of high-performing legacy models is the central driver of the trends observed in our review.

4.2.2 Emerging and niche explanation paradigms

While SHAP and LIME form the methodological core of the field, the review identified several alternative and emerging paradigms that offer different types of explanations, challenging the notion that feature attribution is the only path to interpretability.

Counterfactual Explanations seek to answer the question, "What is the minimal change to the input features that would flip the model's prediction?" This approach is highly intuitive for human users as it provides concrete, actionable examples of how a decision could be different (Hashemi et al. 2025). Research has explored using counterfactuals not only for explanation but also for data augmentation in imbalanced datasets (Temraz and Keane 2022). More recently, large language models have been used to transform counterfactuals into compelling narratives, making them more accessible to non-technical stakeholders (Martens et al. 2025).

Inherently Interpretable Models (or "glass-box" models) represent a different philosophy. Instead of explaining a black box model post-hoc, this approach focuses on designing models that are transparent by construction(Medianovskyi et al. 2023; Dritsas and Trigka 2025). An example from the literature is the Tsetlin Machine, a rule-based model using propositional logic that has achieved competitive accuracy on tasks like fraud detection while remaining fully interpretable, thus avoiding any potential fidelity gap between a model and its explanation (Dritsas and Trigka 2025).

Novel Frameworks are also beginning to emerge. For instance, (Dentamaro et al. 2025) introduced the Evolutionary Independent Deterministic Explanation (EVIDENCE) theory, a deterministic, model-independent method proposed as a more robust and consistent alternative to the heuristic-based approaches of SHAP and LIME. Such work, though nascent, signals a move toward developing more rigorous and specialized XAI techniques. To provide a clear comparative synthesis, the key characteristics of these dominant paradigms, such as feature attribution, counterfactuals, and inherently interpretable models, are summarized in Table 3. This table contrasts the paradigms across critical operational dimensions, includ-

Table 3 Comparative analysis of dominant XAI paradigms in the literature

Paradigm	Core Question Answered	Fidelity	Stability	Actionability	Sensitivity to Imbalance	Corpus Count
Feature Attribution (SHAP, LIME)	How much did each feature contribute to this prediction?	High (by definition, reflects the model's logic)	Potentially low; sensitive to data perturbations and resampling artifacts.	Low; explains "what" but not "how to change."	High; vulnerable to explaining artifacts from SMOTE/GANs.	35
Counterfactuals	What is the smallest change to get a different outcome?	Medium (may find unrealistic feature combinations)	Medium: multiple valid counterfactuals can exist.	High; provides concrete examples for recourse.	Medium: generation can be affected by data density.	5
Inherently Interpretable (Rule-based, Linear)	What is the explicit logic of the model?	Perfect (The model is its own explanation)	High; logic is fixed and transparent.	High; rules are directly actionable.	Low; less prone to artifacts as data is not altered.	7
Methodological/ Survey Papers	N/A	N/A	N/A	N/A	N/A	2

ing their fidelity, stability, and actionability. It also provides a quantitative tally of their prevalence within our corpus, directly addressing the need for a structured, multi-faceted comparison.

4.3 A thematic analysis of application domains (RQ2)

The concentration of XAI research in specific domains is not coincidental; it directly reflects where the drivers for transparency, regulatory pressure, financial risk, and the need for human trust are most acute. This section analyzes why these domains are hotspots for XAI adoption and how their specific challenges connect to the methodological choices discussed previously, thus fitting the pieces of the landscape together. Table 4 details the overview of XAI applications across key domains identified.

4.3.1 Financial services

Financial Services is the predominant application area, with 21 studies, because it exists at the epicentre of the forces driving XAI adoption. The primary drivers include intense regulatory pressure for auditable decisions, the high financial stakes of risk management, and the need to maintain customer trust, which creates a powerful demand for transparency. This directly explains the methodological trend observed in Sect. 4.2: the dominance of a theoretically grounded and defensible method like SHAP is a direct response to the compliance and auditability requirements of the financial sector.

- Credit Card Fraud Detection – Techniques like SHAP and LIME explain models such as Random Forest and Decision Trees, enabling analysts to understand flagged transactions and reduce false positives (Martins et al. 2024; Yousefimehr and Ghatee 2025;

Table 4 Overview of XAI applications across key domains

Domain	Count	Primary Problem(s)	Data Type(s)	XAI Main Drivers	Common Techniques
Financial Services	21	Credit/Loan Default, Credit Card Fraud, Trading Risk, E-commerce	Tabular, Time-Series	Regulatory Compliance, Risk Mitigation, Customer Trust	SHAP, LIME, XGBoost, LightGBM, Autoencoder
Healthcare	7	Medical Claims Fraud, Misdiagnosis, Treatment Efficacy	Tabular, Medical Imaging, Sensor Data	Clinician Trust, Patient Safety, Auditability	SHAP, LIME, CNN, Transformer, Random Forest
Blockchain & Cybersecurity	4	Illicit Crypto Transactions, Smart Contract Fraud	Transaction Graphs, Bytecodes, Network Logs	Establishing Trust in Decentralized/Adversarial Systems	SHAP, 1D CNN, LSTM, Custom Pipelines
Manufacturing & Industrial	3	IIoT Process Monitoring, Clogging Detection, Process Alarms	Sensor Data, Time-Series	Operational Safety, Process Reliability	Autoencoder, GCN, Custom Frameworks
Energy	1	Building Energy Optimization	Tabular, Time-Series	Energy Efficiency, Cost Reduction	Decision Tree, Random Forest, Regression Models
Environment	1	Air Quality Anomaly Detection, Toxicity Assessment	Tabular, Sensor Data	Public Health, Environmental Monitoring	Random Forest, SHAP, SVC
Transportation	1	Flight Delay Prediction	Tabular, Time-Series	Service Reliability, Risk Mitigation	SHAP, Network Analysis
Miscellaneous	11	Proposing/Surveying XAI Methods, User-Centric Evaluation	Varied (Synthetic, Benchmark Datasets)	Advancing XAI Theory, Improving Evaluation	SHAP, LIME, Counterfactuals, Custom Frameworks

Arsenault et al. 2025).

- Credit Risk and Loan Assessment – XGBoost, LightGBM, and SHAP are used to make transparent loan approval decisions and ensure compliance with fair lending laws (Damanik and Liu 2025; Darvish et al. 2025; Lin et al. 2025; Ballegeer et al. 2025).
- Banking and Trading – Autoencoders detect illicit transactions, with SHAP providing interpretable anomaly explanations (Karnavou et al. 2025). In trading, Transformer and LSTM models explain complex market patterns (Sokolovsky et al. 2023; Sasikumar and Raj 2025).

4.3.2 Healthcare and insurance

7 studies address healthcare challenges, focusing on trust, safety, and fraud detection.

- Medical Claims Fraud – Hybrid models (e.g., LSTM + Random Forest) with SHAP-based explanations help auditors validate flagged claims (Lundberg and Lee 2017; Awosika et al. 2024; Mozumder et al. 2025).
- Clinical Decision Support & Diagnosis – CNN, Transformer, and IoT biosensor models explain diagnoses such as cancer and liver disease (Chadaga et al. 2023; Thuy and Benoit 2024; Xu et al. 2025; Kumar et al. 2025).

4.3.3 Blockchain & cybersecurity (4 studies)

This area focuses on establishing trust in decentralized and adversarial systems. Research includes detecting illicit cryptocurrency transactions (Hasan et al. 2024, 2025) and identifying fraudulent smart contracts on the Ethereum blockchain (Ghosh 2025; Jin et al. 2025).

4.3.4 Other domains

Beyond the dominant fields of finance and healthcare, the reviewed studies cover a range of specialized and emerging domains, reflecting the expanding application of XAI. A significant portion of the selected studies is also dedicated to developing general methodologies rather than focusing on a single domain.

- Manufacturing & Industrial (3 studies): In this sector, XAI is applied to enhance operational safety and process reliability. Examples include monitoring Industrial Internet of Things (IIoT) processes (Abudurexiti et al. 2025), explaining process alarms (Bhakte et al. 2023), and detecting tunneling blockages in construction (Gao et al. 2023).
- Niche Domains (1 study each): Several domains are represented by a single, focused study, including Energy, for optimizing building energy consumption (Leuthe et al. 2024); Environment, for assessing air quality and particulate matter risk (Jairi et al. 2025); real estate machine learning models (Acharya et al. 2024)and Transportation, for predicting flight delays (Tan et al. 2025).
- Miscellaneous (11 studies): A substantial number of papers do not target a specific application domain. Instead, their primary contribution is a new XAI technique, a survey, or a general framework. These include foundational tutorials (Bennetot et al. 2025), surveys on explainable anomaly detection (Li et al. 2024), and novel methods for explaining autoencoders (Antwarg et al. 2021) or augmenting data with counterfactuals (Temraz and Keane 2022).

4.4 Critical challenges at the intersection of data, models, and explanations (RQ3 & RQ4)

Beyond mapping the landscape, the thematic synthesis uncovered two profound and interconnected challenges that represent significant hurdles to the reliable deployment of XAI in fraud detection. These challenges, related to data preprocessing and evaluation, are often under-discussed in the literature yet have critical implications for the validity of the generated explanations.

4.4.1 The explainability-imbalance paradox (RQ3)

A common feature of real-world fraud detection datasets is severe class imbalance: fraudulent cases are rare, often making up less than 1% of the data. To prevent models from simply predicting the majority (non-fraudulent) class, researchers use various data-level strategies. The most common are oversampling methods like the Synthetic Minority Over-sampling Technique (SMOTE), which generates new synthetic minority instances, and generative models such as Generative Adversarial Networks (GANs), which learn to produce realistic fraudulent data (Temraz and Keane 2022; Damanik and Liu 2025; Lee et al. 2025).

While these techniques often enhance predictive metrics like recall and F1-score, this review highlights a critical and paradoxical side effect: the process of rebalancing the data can silently undermine the faithfulness of post-hoc explanations. SHAP and LIME are especially vulnerable to this distortion because they compute feature importance using marginal contribution (SHAP) or perturbation-based surrogate modeling (LIME). If the training set is flooded with synthetic samples that distort feature correlations or distributional balance, these methods may assign importance based on patterns that do not exist in real-world data. For example, if SMOTE generates fraudulent transactions with exaggerated amounts or implausible merchant codes, SHAP might overemphasize these as fraud indicators, leading analysts astray. This results in the explainability-imbalance paradox, where the data preprocessing pipeline seems "explanation-agnostic." It is optimized solely for prediction, without regard for its effect on the subsequent explanation task. The causal chain is as follows:

1. A resampling technique (e.g., SMOTE) is applied to the original, imbalanced dataset.
2. This creates a new, balanced training set containing both real and synthetic data points.
3. The black box model is trained on this modified dataset, learning patterns from both genuine and artificial instances. (Guidotti et al. 2019)
4. An XAI method (e.g., SHAP) is then applied to explain the decisions of this trained model.

The resulting explanation can be model-faithful yet world-unfaithful: it reflects patterns learned under an intentionally shifted training distribution but not the operational one. In ML terms, resampling induces covariate shift in $p_{\text{train}}(x)$relative to $p_{\text{ops}}(x)$; explanations then reflect this altered background. Viewed through a bias, variance lens at the explanation layer, resampling often reduces predictive bias toward the majority class while inflating variance of the attribution function (instability across seeds/bootstraps/priors). These dynamics explain why explanation maps may look plausible yet be distribution-dependent and brittle in production. The two case vignettes presented in Box 1 illustrate this paradox in practice.

This critical issue was explicitly examined in a small but significant subset of the reviewed papers. (Cemernek et al. 2024) found that resampling techniques can "jeopardize interpretability," suggesting that cost-sensitive learning (which modifies the model's loss function rather than the data) is a safer alternative for preserving explanation fidelity. In a direct empirical investigation, (Ballegeer et al. 2025) demonstrated that as class imbalance increases, the stability of explanations from both SHAP and LIME decreases significantly, emphasizing a direct trade-off between cost-efficiency gained from certain models and the dependability of their explanations. To address this, explanation-aware training pipelines could be implemented, where resampling techniques are tested for their impact on explanation stability before deployment. Alternatives like cost-sensitive learning, which maintain the original data distribution while adjusting model focus, have shown promise. Another option is explanation filtering, where explanations are generated only for predictions on non-synthetic samples, or using synthetic sample tagging so downstream explanations can account for sample origin. Table 5 presents the class imbalance techniques and their effects on explainability.

Box 1 Case Vignettes Illustrating the Explainability-Imbalance Paradox in Practice

Vignette 1 (Credit Card Fraud)	Vignette 2 (Insurance Claims)
″ A model trained on SMOTE-resampled data flags a transaction. ″ The SHAP explanation shows that “Merchant Category: Electronics” is the top driver of fraud risk. ″ An analyst, seeing this, places all electronics merchants on a watchlist, creating unnecessary friction for legitimate customers and businesses. However, the high importance was an artifact; SMOTE had created many synthetic high-value transactions for this category to balance the dataset. ″ An “explanation-aware” pipeline using cost-sensitive learning would not have altered the data distribution. ″ Its SHAP plot would have been more faithful to reality, correctly highlighting the true drivers (e.g., transaction velocity, time of day) and preventing the misguided action.	″ A LIME explanation for a flagged medical claim highlights “Procedure Code X” as the primary reason for the flag. ″ The analyst, being diligent, perturbs the claim amount by a few cents and re-runs the explanation, which now points to “Provider ID Y,” a completely different reason. ″ This instability reveals that the explanation is an unreliable artifact of the model’s local decision boundary. ″ An “explanation-aware” pipeline would automate this stability stress-test, flagging the explanation as having high variance under minor perturbations and preventing the analyst from acting on spurious and unstable information.

4.4.2 Theoretical foundations of the paradox

The explainability-imbalance paradox is not merely an empirical observation but is theoretically grounded in two foundational machine learning concepts. First, data resampling intentionally induces covariate shift: the training distribution $p_{\text{train}}(x)$ is transformed to differ from the deployment distribution $p_{\text{test}}(x)$. This can improve predictive performance, but also means the model learns from a data reality that does not exist in production. Therefore, we obtain what we define explanation shift: A systematic change in the distribution of explanation values (e.g., SHAP values) when a model is trained on resampled versus original data. In practice, it can be detected by comparing SHAP value distributions for a validation set across the two training procedures (for example, via Kullback–Leibler divergence). Routine monitoring of such explanation shift is an important, yet often overlooked, requirement for robust XAI deployment.

Second, the paradox can be viewed through a bias–variance trade-off at the explanation layer. Resampling methods such as SMOTE reduce predictive bias toward the majority class, but they often increase the variance of the explanation function. Training on synthetic data encourages the model to fit artificial patterns, and post-hoc methods then faithfully explain this overfitting. The resulting feature attributions can be unstable and unreliable when applied to real-world data.

Table 5 Analysis of class imbalance techniques and their reported impact on explainability

Imbalance Technique	Principle	Representative Studies	Reported/Potential Impact on XAI Fidelity	Risk Level
Oversampling (e.g., SMOTE)	Creates synthetic minority samples by interpolating between real neighbors.	(Damanik and Liu 2025; Yousefimehr and Ghatee 2025)	High risk of explaining synthetic artifacts; may generate unrealistic samples that lead to misleading explanations.	**High**
Undersampling	Removes samples from the majority class to balance the dataset.	(Hasan et al. 2024; Yousefimehr and Ghatee 2025)	Can lead to significant information loss, potentially biasing explanations by training the model on an unrepresentative data subset.	**Medium**
Generative Models (e.g., GANs)	Learns the distribution of the minority class to generate new, realistic samples.	(Lee et al. 2025)	Can generate more realistic data than SMOTE, but still carries a high risk of the model learning and the XAI explaining generated artifacts.	**High**
Cost-Sensitive Learning	Penalizes misclassification of the minority class more heavily.	(Cemernek et al. 2024; Ballegeer et al. 2025)	Considered more stable; explanations reflect the true data distribution as the data itself is not modified.	**Low**

4.4.3 Mathematical formulation of explanation distortion

The distortion of explanations by resampling is not merely conceptual; it is a direct consequence of the statistical transformations applied to the data. The Synthetic Minority Over-Sampling Technique (SMOTE) generates a synthetic sample, s, as a linear interpolation between a minority instance, x, and one of its k-nearest minority neighbors, x_R:

$s = x + u \cdot (x_R - x)\,,\ where\ u \sim U\,(0, 1)$

As demonstrated by (Blagus and Lusa 2013), this process has a predictable effect on the feature distribution. While the expected value of the feature remains unchanged $(E(s) = E(x))$, the variance is systematically reduced $\left(\mathrm{var}\,(s) = \frac{2}{3}\mathrm{var}\,(x)\right)$, assuming independence between x and x_R. This mathematical result provides the mechanism for the paradox. Post-hoc explanation methods like SHAP and LIME do not operate in a vacuum; they depend on the statistical properties of the data the model was trained on. SHAP, for instance, computes feature importance by comparing a prediction to a baseline expectation, which is derived from a background distribution of data points. When SMOTE creates an artificially dense, low-variance cluster of minority points, this background distribution becomes unrepresentative of real-world data. Consequently, the marginal contribution of features used in the SMOTE interpolation (e.g., 'Transaction Amount') is systematically overestimated, as the model has learned that small changes in these features within the synthetic cluster lead to large changes in the prediction outcome. The explanation, therefore, becomes faithful to the synthetic data artifacts but unfaithful to the true, underlying fraud patterns.

4.4.4 SHAP vs. LIME under fraud-specific constraints

SHAP and LIME lack reporting with respect to several aspects, such as distribution dependence, stability, fidelity, latency, and gaming risks. Table 6 outlines the potential reporting needed for SHAP and LIME in terms of each aspect to mitigate the risks.

- ***Background dependence (SHAP).*** Per-instance attributions depend on the background/reference used for expectations. Backgrounds derived from resampled data (or mixed time windows) can tilt top-k features and inflate salience for synthetic artifacts. **Action**: fix a production-consistent background and report its construction (time window, class priority, sampling).
- ***Off-manifold perturbations (LIME).*** Local surrogates rely on perturbed samples. In sparse, rule-constrained transaction spaces, many draws are invalid/off-manifold, yielding low local fidelity and flip-prone coefficients sensitive to kernel width/scale. **Action**: constrain perturbations to valid manifolds; report kernel and sampling scheme; verify local fidelity.
- ***Operational cost.*** Per-alert explanations face tight latency budgets (e.g., card-not-present). If only a subset of alerts can be explained, selection must be explicit and auditable.
- ***When post-hoc is not the right tool.*** For policy-grade justifications, mandated monotonicity, or courtroom-style auditability, prefer inherently interpretable models (e.g., monotonic EBM/GAM, rule lists) or hybrid pipelines (interpretable backbone + targeted post-hoc forensics).

Reporting standards for post-hoc explanations in fraud analytics.

- **Provenance**: Declare resampling/weights and do not compute explanations on resampled backgrounds.
- **Background/perturbations**: Specify background set (time span, priors) or perturbation kernel/constraints.
- **Stability**: Bootstrap explanations; report top-k Jaccard and MAD with thresholds.
- **Fidelity**: Include deletion/preservation (or insertion) tests and summarize curves.
- **Latency & coverage**: State per-instance cost and the % of alerts receiving explanations.
- **Role in pipeline**: Diagnostic/monitoring vs. decision-justification; justify when post-

Table 6 Practical risks and mitigations for SHAP and LIME in fraud detection

Aspect	SHAP	LIME	Potential Reporting Needed:
Distribution dependence	Sensitive to background set and class priors; resampled backgrounds bias attributions	Sensitive to perturbation distribution; off-manifold samples degrade fidelity	Background construction (time window, priors), or perturbation scheme & constraints
Stability	Top-k may shift with prior/trees; resampling provenance matters	Coefficients/ top-k can flip across seeds/ kernel width	Bootstrap stability (e.g., top-k Jaccard, MAD); thresholds used
Fidelity	Generally strong for trees; weaker with strong feature dependence	Varies; must be verified locally	Deletion/ preservation or insertion curves; local (R^2) for surrogates
Latency/throughput	Moderate–high; batchable offline	Moderate; sample-dependent	Per-instance runtime and coverage (% alerts explained)
Gaming risk	Stable, predictable attributes can be nudged	Parameter-revealing configs can be gamed	Randomization/ aggregation; post-deployment drift monitoring

hoc suffices vs. interpretable first.

4.4.5 The evaluation vacuum (RQ4)

Another finding of this review is the widespread absence of standardized, rigorous methods for evaluating the quality of the explanations themselves. Foundational XAI research has established that explainability is not a monolithic property but a multi-faceted concept that must be evaluated along several distinct dimensions. However, the applied fraud literature has largely failed to adopt these rigorous frameworks. The dominant practice, observed in nearly 80% of the reviewed studies (39 out of 49; 95% CI [66.3%, 89.1%]), is to use the predictive performance of the underlying ML model (e.g., F1-score, AUC-ROC) as an implicit proxy for the quality of the XAI framework (Damanik and Liu 2025; Dwi Hartomo et al. 2025; Mozumder et al. 2025). This is a flawed assumption, as a highly accurate model can still have incorrect, unstable, or useless explanations. Most post-hoc studies did not report background specifications or stability/fidelity checks, underscoring the evaluation vacuum and motivating the mentioned standards. In contrast, a few studies have begun to pioneer more direct and meaningful evaluation approaches, which can be categorized using formal terminology from the XAI evaluation literature:

- ***Functionally-Grounded Evaluation***: This refers to objective, computable metrics that assess intrinsic properties of the explanation without requiring human experiments. These methods measure properties like Correctness/Fidelity (how truthfully the explanation reflects the model), Stability/Robustness (how consistent explanations are for similar inputs), and Compactness (how simple the explanation is). For example, (Antwarg et al. 2021) quantitatively measured the correctness of SHAP explanations, while(Ballegeer et al. 2025) explicitly measured the stability of SHAP and LIME.
- ***Human-Grounded Evaluation***: This is the rarest but arguably most important type of evaluation, as it measures how the explanation affects a human user. The few studies in our collection that lead this approach provide a blueprint for more detailed work. These studies usually involve specific user roles, like fraud analysts or compliance officers, completing domain-specific tasks such as alert triage or claims validation. The measures used to evaluate the explanation's influence include objective metrics like decision latency and task accuracy, as well as subjective metrics like perceived trust and satisfaction, often gathered through surveys (Leuthe et al. 2024). More sophisticated approaches have even used eye-tracking technology to objectively measure user attention and cognitive load when interacting with explanations (Meza Martínez et al. 2023).

The vast disparity between the common practice of proxy evaluation and these more rigorous methods highlights a critical gap. Without robust evaluation, claims of enhanced trust, transparency, and actionability remain largely unsupported. Future research should operationalize evaluation using existing XAI frameworks to move the field forward. For instance, the taxonomy proposed by (Doshi-Velez and Kim 2017) offers a useful distinction between proxy, functionally grounded, and human-grounded evaluations. Similarly, the FAITH framework (Mohseni et al. 2021) offers a structured set of metrics to assess fidelity, interpretability, and usefulness across different user groups. Incorporating such frameworks can guide the selection of evaluation techniques aligned with the intended users and use cases. A practical next step would be the creation of benchmark datasets specifically annotated for explanation quality evaluation, including user ratings, domain expert feedback, and task performance comparisons.

Beyond technical correctness, the ultimate test of an explanation's value lies in its usability by human stakeholders. Fraud detection environments involve analysts, compliance officers, and regulators, each with different interpretability needs. However, as detailed in Table 7, fewer than 5% of the reviewed studies incorporated user-facing evaluations. This omission represents a serious methodological gap. Mathematically supported explanations may still be incomprehensible, irrelevant, or misleading to practitioners. Future studies should incorporate task-based user studies (e.g., "Can an analyst more accurately resolve alerts with this explanation?") and trust/satisfaction surveys with fraud investigators. Methods such as eye-tracking, think-aloud protocols, or decision-making latency could serve as proxies for cognitive load and interpretability. Without such human-grounded validation, claims of explainability remain uncertain.

Table 7 A taxonomy of evaluation approaches for XAI in fraud detection

Evaluation Category	Core Question	Metrics/ Methods	Frequency in Corpus	Strengths & Weaknesses	Representative Studies
Proxy Evaluation	Is the underlying model accurate?	AUC, F1-Score, Precision, Recall	Very High (~80%)	Weakness: Indirect and flawed. An accurate model can have incorrect, unstable, or useless explanations.	(Damanik and Liu 2025; Mozumder et al. 2025)
Functionally Grounded (Quantitative)	Does the explanation have desirable formal properties?	Correctness/ Fidelity: Deletion/ Preservation Checks. Stability: Explanation similarity on perturbed inputs. Compactness: Number of features.	Low (~10–15%)	Strength: Objective, scalable, and does not require users. Weakness: Does not guarantee the explanation is actually useful to a human.	(Antwarg et al. 2021; Arcudi et al. 2024; Ballegeer et al. 2025)
Human-Grounded (User-Perceived)	Does the explanation help the human user?	Understandability: User comprehension tests. Usefulness: Task performance improvement. Trust: User satisfaction surveys, eye-tracking.	Very Low (<5%)	Strength: The ultimate test of an explanation's value. Weakness: Subjective, expensive, hard to scale, risk of bias.	(Meza Martínez et al. 2023; Leuthe et al. 2024)

5 Discussion

This systematic review synthesizes the current state of XAI in fraud detection, highlighting a field marked by rapid growth, strong potential, and important foundational challenges. The findings indicate a domain actively using available tools to address pressing real-world challenges but has not yet established the mature methodologies and evaluation standards needed for broad, reliable implementation. This discussion interprets the main findings, identifies research gaps, and outlines a specific research plan to advance the field. Although much of the reviewed literature focuses on developments from 2021 onward, the core methodological issues, like fidelity, stability, and user trust, were addressed in foundational work and remain highly relevant to the gaps identified in this review.

5.1 Interpreting the findings: beyond the methodological (RQ5)

Our review highlights several key themes. The prominence of SHAP and LIME indicates a methodological monoculture, likely driven by their accessibility and model-agnostic design. It allows them to be applied to existing high-performing fraud detection systems without altering the architecture. This over-reliance may limit innovation in explanation methods better suited for fraud detection. For instance, counterfactual explanations provide actionable insights (e.g., "Had the claim amount been lower, it would not be flagged") that match the analysis approach of analysts. Likewise, causal models, being computationally demanding, can produce explanations that resist forged correlations and adversarial inputs. Future research should focus on integrating causal inference frameworks (e.g., structural causal models, invariant risk minimization) into fraud detection workflows. Moreover, hybrid approaches combining rule-based interpretable models (e.g., Tsetlin Machines, symbolic systems) with feature attribution layers could deliver both accuracy and transparency. Moving beyond the monoculture requires shifting from convenience to contextual appropriateness, such as selecting explanation types based on domain needs rather than popularity or ease of implementation.

The findings on class imbalance expose a hidden risk in typical data science workflows. Relying on synthetic resampling to boost predictive metrics without also checking how it affects explanation accuracy is a critical methodological flaw. This can result in analysts and regulators making decisions based on explanations that stem from synthetic data artifacts rather than actual fraud patterns, which weakens the trust that XAI is supposed to establish (Sokolovsky et al. 2023; Cemernek et al. 2024; Damanik and Liu 2025). This phenomenon is a clear example of a "data cascade," where traditional data processing practices trigger negative downstream effects. As described by (Sambasivan et al. 2021), such cascades often go unnoticed until late in the development lifecycle. The paradox is therefore not an isolated XAI problem but a critical data-centric issue where an early pipeline decision, applying SMOTE, cascades downstream to corrupt the final output meant to ensure accountability.

Without reliable methods to measure explanation quality, claims of increased trust and transparency remain scientifically unproven. This lack of thorough evaluation is a key obstacle to the field's development (Meza Martínez et al. 2023; Xu et al. 2025; Ballegeer et al. 2025). Our findings suggest that the applied field has largely failed to heed the foundational calls for a more rigorous science of evaluation issued nearly a decade ago (Doshi-Velez and Kim 2017). To make XAI trustworthy, the focus must shift from merely generating explanations to systematically validating them. This has significant implications for both theory and practice. Theoretically, it challenges the field to develop a stronger science of explanation evaluation. Practically, it warns regulators and institutions that deploying an "explainable" system without a validated evaluation framework only creates an illusion of transparency (Awosika et al. 2024). To translate these principles into a coherent operational workflow, we propose a conceptual model of an "Explanation-Aware Fraud Detection Pipeline," as illustrated in Fig. 6. This model moves beyond the linear 'explain-as-an-afterthought' approach and integrates validation checkpoints throughout the model lifecycle.

This pipeline integrates imbalance checkpoints and evaluation gates directly into the workflow. It consists of four main stages: (1) Data Preprocessing, which includes an "Imbalance Checkpoint" to test the impact of resampling on explanation stability; (2) Model Training, which incorporates explanation-aware techniques; (3) Post-Hoc Explanation, where explanations are generated; and (4) Validation & Deployment, which features two "Evaluation Gates"—a quantitative gate for functional checks (fidelity, stability) and a qualitative gate for human-grounded evaluation before deployment.

We have developed a pre-deployment validation checklist to translate our evaluation taxonomy into a practical tool for practitioners. As detailed in Fig. 7, this checklist converts the core principles of our findings, such as fidelity, stability, data provenance, and human usefulness, into a set of concrete validation actions with clear, measurable pass criteria. This provides a structured framework to help organizations ensure that their XAI systems are robust, reliable, and auditable before being deployed in high-stakes environments.

5.2 Addressing the critical gaps: a proposed research agenda (RQ5)

Based on the comprehensive analysis, this review identifies four critical research gaps that must be addressed. These gaps, summarized in Table 8, form the basis of a proposed research agenda.

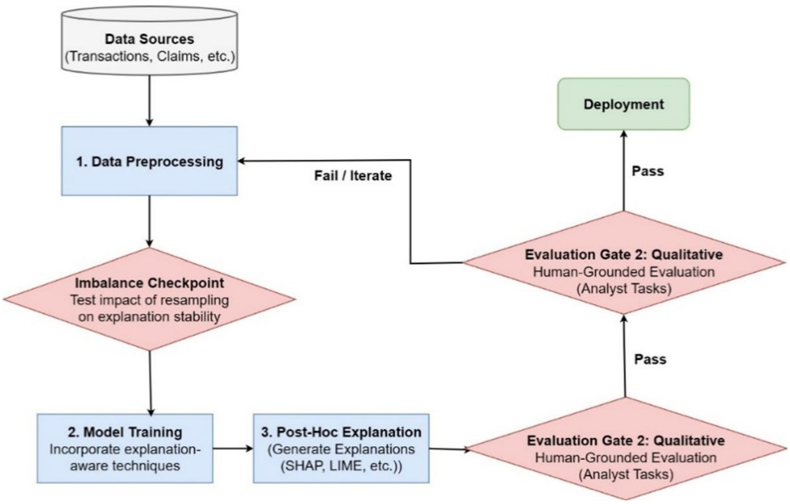

Fig. 6 A conceptual framework of an explanation-aware fraud detection pipeline

Pre-deployment XAI Validation Checklist for Fraud Analytics

Check	Dimension • Validation Actions	Pass Criteria
☐	**Fidelity — Does the explanation reflect the model's logic?** • Run deletion/preservation tests (remove top features, verify expected performance drop). • Perturb key features; check monotonic response. • Counterfactual alignment (minimal changes flip outcome consistently).	• Fidelity threshold met (e.g., ≥80% feature–performance alignment). • Deletion/preservation curves show expected order. • Counterfactuals are plausible and consistent.
☐	**Stability — Are explanations robust to small changes?** • Bootstrap or K-fold: recompute explanations across resamples. • Small input noise/perturbations; track explanation similarity. • Report similarity (cosine/Jaccard) for top-k features.	• Stability threshold met (e.g., similarity ≥0.70 across runs). • Rank order of top features stable (e.g., ≥80% overlap).
☐	**Provenance — Do explanations account for data origin?** • Tag case provenance: real vs synthetic (SMOTE/GAN/etc.). • Log all preprocessing and imbalance treatments. • Perform "synthetic-artifact" audit: do top attributions depend on synthetic-only patterns?	• Explanations for deployable cases are based on real-data patterns. • Synthetic-only drivers flagged/filtered. • Full provenance and preprocessing documented.
☐	**Human usefulness — Do explanations help analysts?** • A/B with analysts: alerts with vs without explanations. • Measure alert-triage accuracy, decision latency, confidence. • Optional: prioritization accuracy; justification quality.	• Measurable benefit vs baseline (e.g., +10% triage accuracy or reduced latency). • Trust not degraded; confidence appropriately calibrated.
☐	**Documentation — Is the system auditable and transparent?** • Disclose peer-review status of methods and datasets. • Provide limitations (timeframe, language, grey literature). • Maintain end-to-end audit trail for regulators (data, model, explanation, decisions).	• Documentation sufficient for regulatory reporting. • Reproducible pipeline (data, preprocessing, model, XAI, human decisions). • Clear statement of limits and risks.

Fig. 7 A proposed pre-deployment XAI validation checklist for fraud analytics

1. ***The Evaluation Gap.*** The most urgent need is to go beyond proxy metrics and create a standardized, multi-faceted framework for evaluating explanation quality, including assessing fidelity, robustness, and human-centric utility.
2. ***The Data-Centric Gap.*** The interplay between data preprocessing and explainability must become a primary research focus, leading to the development of "explanation-aware" data processing pipelines.
3. ***The Human-Centric Gap.*** There is a noticeable lack of user studies involving the actual stakeholders of fraud detection systems. Research should shift from a purely algorithmic approach to focusing on a human-computer interaction perspective.
4. ***The Methodological Gap.*** To break the current monoculture, research should actively explore and compare a broader range of explanation paradigms, especially those based on causality and counterfactual reasoning.

To address the critical gap in human-centered evaluation, we propose a novel, ecologically valid task rubric for assessing the practical utility of XAI systems in fraud analytics. As detailed in Fig. 8, it moves beyond generic measures by mapping specific evaluation tasks and metrics to the real-world workflow of a fraud investigation, from initial alert triage to final compliance review. This provides a concrete and standardized framework for future research to generate more meaningful and comparable results.

5.3 Operational challenges: adversarial robustness and computational latency

Beyond data and evaluation, two operational challenges threaten the real-world deployment of XAI in fraud detection:

1. ***Adversarial Robustness***: Recent research indicates that post-hoc explanation methods are vulnerable to adversarial attacks. (Slack et al. 2020) demonstrated a "scaffolding" technique where a malicious actor can craft a classifier that behaves in a biased way on

Table 8 Summary of identified research gaps and future directions

Identified gap	Description	Proposed future direction
The evaluation vacuum	The lack of standardized metrics and protocols for evaluating explanation quality.	Develop and adopt standardized evaluation protocols based on frameworks like (Doshi-Velez and Kim 2017) and FAITH (Mohseni et al. 2021). Establish benchmark datasets where transactions are annotated not only with fraud labels but also with ground-truth explanations or key contributing features validated by human experts.
The explainability-imbalance paradox	The common use of data resampling techniques can undermine explanation fidelity.	Design "explanation-aware" data processing and cost-sensitive learning methods.
The human-centric gap	Almost no studies involve fraud analysts or compliance professionals in evaluating explanations, despite their central role in operational decisions.	Design domain-specific, task-based user studies to assess whether XAI improves accuracy, decision speed, and trust. Incorporate behavioral metrics (e.g., error rates, attention tracking) alongside subjective measures.
The methodological monoculture	SHAP and LIME dominate the literature due to their accessibility and model-agnostic design, but are often used uncritically, even when better-suited paradigms exist.	Promote comparative research on causal, counterfactual, and inherently interpretable models. Encourage hybrid and domain-adaptive approaches that address fraud-specific explanation needs (e.g., recourse, root cause analysis, adversarial robustness).

real data but is designed to produce harmless-looking explanations when examined with perturbation-based methods like LIME and SHAP. This occurs because these methods generate explanations based on the model's behavior on out-of-distribution samples, which the scaffolded model is trained to handle differently. This underscores the urgent need for developing robust explanations that are resistant to such manipulation.

2. ***Computational Latency***: The computational overhead of methods like KernelSHAP can be prohibitive for real-time fraud detection systems that need millisecond-level responses. Although this is a known issue, promising mitigation strategies are actively being researched. These include developing highly efficient approximate SHAP algorithms (e.g., FastSHAP) and using distilled, model-specific surrogate models that can generate explanations much more quickly in production environments.

5.4 Limitations and threats to validity

This review has several limitations. First, restricting sources to peer-reviewed journal articles in English (2021–2025) may introduce publication bias and language bias; relevant

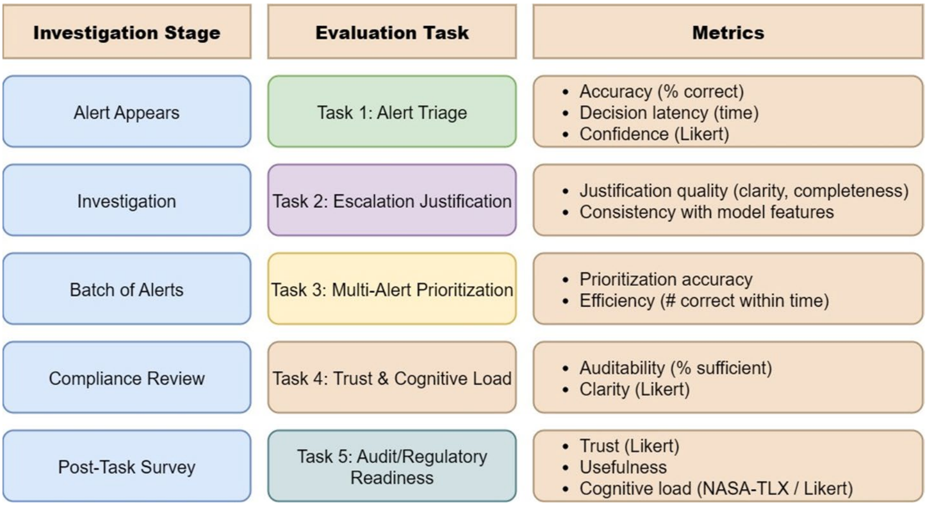

Fig. 8 A proposed human-grounded evaluation rubric for fraud analytics

findings in non-English outlets or grey literature (e.g., industry reports, dissertations) were excluded by design. Second, selection bias may persist despite dual screening and a high inter-rater agreement, because judgments about "substantive XAI use" sometimes require interpretation of ambiguous reporting. Third, the corpus is heterogeneous across domains (banking, insurance, e-commerce, AML/crypto) and data regimes; therefore, prevalence estimates (e.g., evaluation practices) should be interpreted as descriptive, not causal, patterns. Fourth, because the literature in XAI and fraud detection is fast-moving, our fixed search window creates time-lag effects: recently accepted articles or in-press errata may not be captured even though we report the exact per-database search dates. Finally, we adapted a domain-appropriate risk-of-bias rubric rather than a single standardized tool; while this improves face validity for XAI/fraud contexts, it can limit direct comparability to other SLRs. We mitigate these concerns by releasing the row-level extraction dataset, codebook, screening logs, and analysis notebooks for full reproducibility and re-analysis.

5.5 Conclusion and future work

This review critically examines the methodological landscape of XAI in fraud detection, highlighting two systemic flaws that threaten the validity of current research: the "explainability-imbalance paradox," where data resampling undermines explanation fidelity, and a significant "evaluation vacuum," where explanation quality is rarely assessed directly. Our analysis of 49 studies reveals a field in need of greater methodological rigor. It provides a detailed distribution of application domains, showing that Financial Services is the most studied area (21 studies), followed by healthcare (7 studies). Many studies are Methodological in nature (11), with smaller groups in Blockchain & Cybersecurity (4), Manufacturing & Industrial (3), and single studies in Energy, Environment, and Transportation. This diversity indicates that while XAI research in fraud detection remains concentrated in finance, it is gradually expanding into other high-stakes sectors. SHAP and LIME continue to be the most widely used explanation methods, with decision trees, gradient boosting models, neural networks, and autoencoders forming the core of the predictive models. Less common but notable are advanced architectures such as transformers, attention mechanisms, and deep generative models. Handling class imbalance remains a persistent challenge, with strategies like oversampling (e.g., SMOTE), undersampling, cost-sensitive learning, and generative augmentation affecting explanation fidelity in different ways. Evaluation practices tend to focus on proxy model performance metrics, with relatively few studies directly measuring explanation quality or involving human-centred evaluations. This reveals a gap between algorithmic transparency and practical interpretability for end-users. Future research should address several open challenges. First, evaluation methods need to evolve to include balanced frameworks that assess performance, explanation quality, and human usability. Second, diversification across domains should be encouraged, especially in emerging sectors like energy, transportation, and environmental monitoring, where AI interpretability can have societal and regulatory impacts. Third, hybrid XAI approaches combining local and global interpretability, as well as model-specific and model-agnostic techniques, could enhance explanation robustness. Lastly, more focus should be placed on operationalizing XAI in real-world decision-making processes, ensuring explanations are not only technically accurate but also actionable, trustworthy, and tailored to stakeholders' needs.

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Data availability The row-level extraction dataset, the data-extraction codebook, screening log, including a list of all full-text articles excluded with reasons, and executable analysis notebooks are available at OSF: [https://osf.io/9nykd/].

Declarations

Conflict of interest The authors have no relevant financial or non-financial competing interests to disclose. No funding was received to support this study.

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